

GIS-I

LECTURE HANDOUT

Chapter One: Introduction to Geographic Information Systems

For BSc in Surveying Engineering Students

Item	Description
Course	GIS-I
Chapter	Chapter One
Coverage	Introduction, Definitions and Terminology, Evolution, Elements and Capabilities, Major Applications
Orientation	Conceptual foundation with practical Surveying Engineering examples

Learning Objectives / Competences

After completing this chapter, students should be able to:

- Be familiar with the basic concepts, objectives, and trends of Geographic Information System (GIS) technology.
- Know GIS capability and applications in decision-making.
- Learn the basic steps to create and maintain GIS data, perform GIS analysis, and present results in hard-copy maps or digital form.
- Understand the definition, history, and application of GIS.
- Relate and identify GIS data sources and integrate them with attribute representations.
- Understand GIS data models and apply them practically.
- Analyze differences between vector and raster data models.
- Collect data from the environment and apply theoretical concepts practically.

Course Description – Chapter Coverage

This chapter introduces the definition, history and development, objectives, components, capabilities, data types, and applications of GIS. The broader GIS-I course also introduces data input, storage and management, digitization, data editing, georeferencing, error detection and removal, GIS data models, data integration, attribute data, relations and joins, queries, shapefile/coverage/layer concepts, map layout, exporting data, and an overview of GIS software.

1.1 Introduction

Geographic Information Systems (GIS) are systems used to collect, store, manage, analyze, visualize, and communicate information that has a geographic or spatial reference.

GIS is more than computerized map drawing. A conventional map mainly communicates geographic location and relationships, while GIS combines geographic information with databases and analytical tools. For example, a road layer can be connected to information about road name, class, surface type, length, and condition.

In Surveying Engineering, field observations such as control points, roads, buildings, drainage structures, parcels, and other features can be collected and organized as GIS layers. GIS can then be used to edit, query, analyze, measure, visualize, and communicate the information.

Key idea: The strength of GIS comes from integrating location with descriptive information and using that information for analysis and decision-making.

1.2 Definition of GIS and Related Terminology

1.2.1 Definition of GIS

A Geographic Information System can be understood as an organized combination of hardware, software, data, people, and procedures designed to capture, store, manage, manipulate, analyze, and display geographically referenced information.

Different authors emphasize different aspects of GIS. Some emphasize the computer system, some the spatial database, some analysis and decision support, and others the integration of technology, people, and procedures. In practical terms, GIS connects spatial location with descriptive information and provides tools for managing and analyzing that information.

1.2.2 Spatial / Geospatial Data

Spatial data describe the location, shape, size, and spatial relationships of geographic features. They are normally referenced to a coordinate system.

- Point: a feature represented by a location, such as a survey control point, water point, or borehole.
- Line: a linear feature such as a road, stream, pipeline, or boundary.
- Polygon: an area feature such as a parcel, lake, administrative unit, or land-use area.

1.2.3 Vector Data

Vector data represent geographic features using points, lines, and polygons. Each feature can be linked to attribute information.

Example: A cadastral parcel can be represented as a polygon, with attributes such as Parcel_ID, area, land-use type, and administrative unit.

1.2.4 Raster Data

Raster data represent geographic space as a regular grid of cells or pixels. Each cell stores a value. Raster data are especially useful for continuous phenomena and remotely sensed imagery.

Examples include Digital Elevation Models (DEMs), satellite images, rainfall surfaces, temperature surfaces, and vegetation-index images.

1.2.5 Vector and Raster Comparison

Aspect	Vector	Raster
Representation	Points, lines and polygons	Grid cells / pixels
Best suited for	Discrete features and boundaries	Continuous surfaces and imagery
Examples	Roads, parcels, buildings	DEM, satellite image, rainfall
Information structure	Attributes linked to features	Values stored in cells
Typical analysis	Network, proximity, polygon overlay	Surface, terrain and image analysis

1.2.6 Attribute Data

Attribute data are descriptive, non-spatial information associated with spatial features. A road, parcel, building, or water point may have many descriptive fields.

- Road: Road_ID, road name, road class, surface type, length, condition.
- Building: Building_ID, use, number of floors, construction type.
- Parcel: Parcel_ID, area, land-use type, administrative unit.

1.2.7 Spatial and Attribute Data Relationship

Spatial data answer questions such as 'Where is it?' Attribute data answer questions such as 'What is it?' or 'What are its characteristics?' GIS connects the two so that users can perform both location-based and attribute-based queries.

1.2.8 Georeferencing

Georeferencing is the process of assigning geographic coordinates and an appropriate coordinate reference system to data that do not have a usable spatial reference. It is commonly applied to scanned maps, aerial photographs, plans, and images.

Example: A scanned map can be georeferenced using identifiable ground-control points whose coordinates are known. The georeferenced map can then be overlaid with other GIS layers.

1.2.9 Topology

Topology describes important spatial relationships such as connectivity, adjacency, and containment. Topological rules can help identify gaps, overlaps, disconnected lines, and other geometric or relational problems.

1.2.10 Spatial Query

A spatial query asks a question using geographic relationships. Examples include identifying buildings within a specified distance of a road or selecting parcels inside an administrative boundary.

1.2.11 Overlay Analysis

Overlay analysis combines two or more spatial datasets to derive new information. A site-selection project, for example, may overlay slope, road access, land use, drainage, and settlement layers.

1.2.12 Buffer

A buffer creates a zone at a specified distance around a point, line, or polygon. For example, a 500 m buffer around a road can be used to identify nearby buildings or areas within a specified distance of the road.

1.3 Evolution of GIS

GIS has developed from traditional cartographic and spatial-analysis practices into integrated digital systems connected to GNSS, remote sensing, UAVs, LiDAR, web technologies, cloud computing, mobile devices, 3D visualization, and automated analytical methods.

1.3.1 Traditional Mapping and Manual Analysis

Before digital GIS, spatial information was mainly represented using paper maps, plans, photographs, and survey drawings. Analysis relied heavily on manual measurement, transparent overlays, drafting, calculation, and professional interpretation.

1.3.2 Early Computerized GIS

The development of computers made it possible to store coordinates and attribute information digitally and automate selected mapping and analysis tasks. Early GIS applications were strongly associated with government planning, land management, natural resources, and large spatial databases.

1.3.3 Desktop GIS

Desktop GIS brought spatial data management, editing, analysis, cartography, and database functions to personal computers. This made GIS practical for universities, engineering offices, planning organizations, survey institutions, and many other users.

1.3.4 Modern GIS

Modern GIS increasingly integrates multiple technologies and data sources, including GNSS, satellite remote sensing, UAV/drone imagery, LiDAR, mobile GIS, Web GIS, cloud services, 3D GIS, real-time data, and AI/machine learning.

- GNSS provides geographic positioning and coordinates.
- Remote sensing supplies satellite and airborne observations.
- UAVs provide high-resolution local imagery and photogrammetric products.
- LiDAR provides dense three-dimensional measurements.
- Web GIS allows maps and services to be accessed through networks.
- Cloud computing supports large datasets and collaborative workflows.
- 3D GIS supports visualization and analysis of terrain and built environments.

- AI and machine learning can support classification, feature extraction, prediction, and pattern recognition.

1.4 Elements of GIS and Capabilities of GIS

1.4.1 Elements / Components of GIS

A functional GIS depends on several interconnected elements. A common framework includes hardware, software, data, people, and procedures.

Component	Description
Hardware	Computers, servers, GNSS receivers, scanners, digitizers, printers, plotters, storage and network equipment.
Software	Programs used for data input, editing, management, analysis, visualization, geoprocessing and mapping.
Data	Spatial and descriptive information collected from surveys, GNSS, maps, imagery, UAVs, LiDAR, databases and other sources.
People	Surveyors, GIS analysts, engineers, planners, technicians, researchers, managers and decision-makers.
Procedures	Methods, standards, workflows, quality-control rules and organizational practices used to produce reliable GIS results.

1.4.2 Hardware and Peripherals

- Input devices: keyboard, mouse, scanner, digitizer, GNSS receiver, total station, and field data collector.
- Processing and storage: workstation, server, external storage, and network storage.
- Display: monitor and other visualization devices.
- Output devices: printer and plotter for maps and reports.
- Communication equipment: network and internet infrastructure.

1.4.3 GIS Software

GIS software provides tools for creating, editing, managing, analyzing, and displaying geographic information. In an ArcGIS Desktop learning environment, ArcMap is used for mapping, editing and analysis; ArcCatalog is used for browsing and managing GIS data; and ArcToolbox provides geoprocessing tools.

1.4.4 Major GIS Capabilities

Capability	Basic purpose
Data input	Enter or import spatial and attribute information.
Data integration	Combine datasets from different sources.
Digitization and scanning	Convert map or image information into digital GIS data.
Georeferencing	Assign an appropriate spatial reference to unreferenced maps or images.
Storage and management	Organize spatial datasets, tables, metadata and related files.
Editing and maintenance	Correct geometry and attributes and update datasets.
Spatial query	Select features according to location or spatial relationship.
Attribute query	Select records according to descriptive fields.
Measurement	Calculate distance, area, length and other spatial quantities.
Buffer / proximity	Identify areas within specified distances.

Overlay	Combine layers to identify spatial relationships.
Network analysis	Analyze connected systems such as roads and utilities.
Visualization	Display geographic information through maps and other graphics.
Map layout and export	Prepare maps for printing, reporting and digital distribution.

1.4.5 GIS Data Input

- Field survey and GNSS measurements.
- Total station and other surveying instruments.
- Scanning and georeferencing of existing maps.
- Digitizing points, lines, and polygons.
- Importing satellite imagery and raster datasets.
- Importing CAD, tabular, GPS/GNSS, and compatible datasets.
- Entering or importing attribute records.

1.4.6 Data Storage and Management

GIS data management involves organizing spatial features, attribute tables, coordinate-reference information, metadata, and related files. Good data management improves accessibility, consistency, updating, backup, and quality control.

1.4.7 Data Editing and Error Detection

Editing is necessary because GIS datasets may contain positional, attribute, topological, or other errors.

- Missing features or attributes.
- Duplicate features.
- Overshoots and undershoots in line data.
- Gaps or overlaps between polygons.
- Incorrect or inconsistent attribute values.
- Incorrect coordinate reference system or georeferencing.

1.4.8 GIS Data Integration Operations

Operation	Basic purpose
Union	Combines polygon datasets and preserves the input polygon areas and their attributes.
Clip	Extracts features or portions of features within a specified boundary.
Merge	Combines datasets of the same general geometry/type into one dataset.
Dissolve	Combines neighboring features based on a common attribute and removes internal boundaries where appropriate.
Edge matching	Matches features along adjoining dataset boundaries.
Intersection	Produces spatial portions where input features overlap and carries relevant attributes.

1.4.9 Non-Spatial / Attribute Data Models

Attribute data are usually organized in tables. Each row represents a record or feature and each column represents a field. A unique identifier can connect the spatial feature to its descriptive record.

1.4.10 Relation Fields and Joining

A join connects records between tables using a common field. The common field should identify corresponding records consistently. For example, Parcel_ID can connect a parcel layer with a separate property or valuation table.

1.4.11 GIS Query

- Spatial query: selects features according to location or spatial relationship.
- Non-spatial/attribute query: selects records according to attribute values.
- SQL query: uses Structured Query Language expressions to select records in supported database environments.

1.4.12 Shapefile, Coverage, and Layer

A shapefile is a widely used vector data format associated with geographic features and an attribute table. A shapefile dataset is composed of several related files. A coverage is an older GIS data model/format historically associated with ArcInfo workflows and topological vector data. A layer is a conceptual representation of a dataset in a GIS map and includes how that dataset is displayed.

1.4.13 Map Layout Compilation

Map layout is the arrangement of map elements into a clear and communicative composition. A professional map commonly includes a title, map body, legend, scale, north arrow, coordinate or grid information where appropriate, data source, author or institution, date, and other necessary marginal information.

1.4.14 Exporting Data and Maps

GIS outputs can be exported in different formats according to their purpose. Spatial data may be exported to compatible GIS formats, while maps may be exported to image or document formats for reporting, printing, presentation, or digital sharing.

1.4.15 Overview of Common GIS Software

Software / platform	General role
ArcView	An earlier desktop GIS environment associated with ESRI and the development of desktop GIS practice.
ArcGIS	A family of GIS software and technologies for mapping, data management, analysis, geoprocessing and visualization.
Idrisi	A GIS and remote-sensing software environment associated strongly with raster analysis and image processing.
QGIS	An open-source desktop GIS used for spatial data management, mapping, analysis and integration.

1.5 Major Application Areas of GIS

Application area	Examples of GIS use
Natural resources and environment	Land-cover mapping, watershed analysis, forest management, soil mapping, erosion studies and environmental monitoring.
Urban and regional planning	Land-use planning, settlement analysis, zoning, infrastructure planning and service accessibility.
Surveying and engineering	Survey-data integration, mapping, asset inventories, site analysis, infrastructure management and project planning.
Transportation	Road-network mapping, route analysis, accessibility, accident analysis, transport planning and maintenance.

Emergency and disaster management	Hazard mapping, exposure assessment, evacuation planning and emergency-resource mapping.
Public health	Health-facility mapping, disease mapping, population analysis and service-area studies.
Agriculture	Land suitability, crop monitoring, soil mapping, irrigation planning and precision agriculture.
Telecommunications	Network planning, tower/site analysis, coverage mapping and infrastructure management.
Mining and geotechnical work	Integration of geology, terrain, mineral information, infrastructure and environmental constraints.
Land administration and property	Cadastral mapping, parcel management, land-use records and valuation-related mapping.
Tourism and recreation	Tourism-resource mapping, route planning, accessibility and attraction inventories.
Education and research	Spatial research, field-data management, mapping, spatial statistics and environmental studies.

Relationship of GIS to Other Geospatial Fields

Field / technology	Relationship to GIS
Surveying	Provides accurate field measurements, coordinates, control points and engineering observations.
GNSS	Provides geographic positioning and coordinate observations.
Remote Sensing	Provides satellite and airborne imagery and measurements.
Photogrammetry	Produces maps, orthophotos, point clouds and 3D information.
Cartography	Provides principles for clear geographic representation and map communication.
Database management	Supports organized storage, retrieval, relationships and queries.
Spatial statistics	Provides methods for identifying spatial patterns and relationships.
UAV / drone	Provides high-resolution imagery and photogrammetric products.
LiDAR	Provides dense three-dimensional measurements.
Programming	Supports automation, customization and reproducible GIS workflows.
AI / Machine Learning	Can support classification, feature extraction, prediction and pattern recognition.

Basic GIS Workflow

1. Define the problem, objectives, study area, and required outputs.
2. Identify and collect appropriate spatial and attribute data.
3. Check coordinate reference systems, data quality, completeness, and source information.
4. Input, import, digitize, scan, or georeference the required data.
5. Create and organize spatial layers and attribute tables.
6. Edit geometry and attributes and detect/remove relevant errors.
7. Integrate datasets using appropriate operations such as clip, merge, dissolve, union, or intersection.
8. Perform queries, measurements, spatial analysis, and other required operations.
9. Visualize results and prepare a suitable map layout.

10. Export the map/data and communicate the findings.
11. Document data sources, methods, assumptions, and limitations.

Worked Conceptual Example: Selecting a Suitable Health-Center Site

Suppose a local administration wants to identify potential locations for a new health center. GIS can support the study by combining several spatial criteria.

12. Collect road, settlement, existing health-facility, land-use, drainage, slope, and administrative-boundary data.
13. Check and standardize the coordinate reference systems.
14. Create buffers around roads if accessibility is an important criterion.
15. Identify areas within or near settlements according to the service objective.
16. Exclude unsuitable areas such as restricted zones, water bodies, or specified environmental constraints.
17. Use elevation or slope information to identify terrain that meets the project requirements.
18. Overlay relevant criteria and identify areas satisfying the defined conditions.
19. Review candidate areas using field verification and additional information.
20. Prepare a final map and supporting table/report.

Important: GIS provides spatial evidence and analytical information. Final decisions may also require social, economic, legal, environmental, engineering, and institutional considerations.

Field Data Collection and GIS

Surveying students should understand that GIS analysis is strongly influenced by the quality of input data. Field data may be collected using GNSS receivers, total stations, leveling instruments, mobile data-collection tools, and other appropriate methods.

- Define what features need to be collected before fieldwork.
- Prepare an appropriate coordinate reference system and control strategy.
- Collect accurate coordinates and descriptive attributes.
- Record relevant metadata such as date, instrument, operator, method, and accuracy.
- Check field observations for completeness and obvious errors.
- Import or digitize the data into the GIS.
- Validate the resulting GIS layers against field conditions where possible.

Key Points to Remember

- GIS integrates spatial location with descriptive information.
- GIS is more than computerized map drawing; it supports data management, querying, analysis, visualization, and decision support.
- The main vector geometry types are point, line, and polygon.
- Raster data represent space using a grid of cells or pixels.
- A GIS commonly involves hardware, software, data, people, and procedures.
- GIS capabilities include input, storage, editing, querying, analysis, visualization, and output.
- Georeferencing is important when spatial data do not have an appropriate geographic reference.
- Attribute tables provide descriptive information associated with spatial features.
- Data integration and overlay allow information from multiple layers to be combined.
- Good GIS analysis depends on appropriate data, correct coordinate systems, data quality, and clearly defined objectives.

- Surveying, GNSS, remote sensing, photogrammetry, cartography, databases, and GIS are complementary parts of modern geospatial practice.

Review Questions

21. Define GIS in your own words and explain why GIS is more than computerized map drawing.
22. What is the difference between spatial data and attribute data?
23. Explain point, line, and polygon vector data with one Surveying Engineering example for each.
24. What is raster data? Give three examples of raster datasets.
25. Compare vector and raster data models.
26. Describe the major stages in the evolution of GIS.
27. List and explain the main elements/components of a GIS.
28. What are the major capabilities of GIS?
29. What is georeferencing and why is it important?
30. Explain topology and give two examples of spatial errors that topology can help identify.
31. Differentiate spatial query from non-spatial/attribute query.
32. What is overlay analysis? Give one practical example.
33. Explain the purpose of buffer analysis.
34. Differentiate union, clip, merge, dissolve, edge matching, and intersection.
35. What is a join and why is a common key field important?
36. Explain shapefile, coverage, and layer.
37. List at least five major application areas of GIS.
38. Explain the relationship between GIS and GNSS, remote sensing, and surveying.
39. Describe a GIS workflow for a simple site-selection problem.
40. Why is data quality important in GIS?

Suggested Practical Exercises

Exercise 1: Identify GIS Components

Select a small GIS project and identify the hardware, software, data, people, and procedures required.

Exercise 2: Vector Feature Identification

Using a local map, identify point, line, and polygon features and explain why each geometry is appropriate.

Exercise 3: Attribute Table Design

Create an attribute table for roads, buildings, or parcels. Identify a unique ID field and at least five descriptive fields.

Exercise 4: Georeferencing Concept

Select a scanned map and identify possible ground-control points. Explain how their known coordinates could be used for georeferencing.

Exercise 5: Buffer Analysis

Create a conceptual 500 m buffer around a road and identify mapped features inside the buffer.

Exercise 6: Spatial and Attribute Queries

Develop two spatial queries and two attribute queries for a local GIS dataset.

Exercise 7: Data Integration

Explain how clip, merge, dissolve, union, and intersection could be used in a land-use or infrastructure project.

Exercise 8: Map Layout

Prepare a map layout containing a title, legend, scale, north arrow, source, date, and other necessary marginal information.

Chapter Summary

GIS is an integrated approach for working with geographic information. It brings together spatial data, descriptive information, technology, people, and procedures to support data capture, storage, editing, analysis, visualization, and communication. Understanding vector and raster data, geometry and attributes, GIS components, and GIS capabilities provides the foundation for later GIS-I topics such as digitization, georeferencing, editing, data integration, spatial queries, spatial analysis, and map production.

References / Suggested Reading

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End of Chapter One – GIS-I